

CLASSIFICATION OF ELEMENTS AND PERIODICITY IN PROPERTIES

21. In the modern periodic table, one of the following does not have an appropriate position:

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- (a) transition elements
- (b) inert gases
- (c) inner transition elements
- (d) halogens

22. Element E belongs to the 4th period and group 16 of the periodic table. The valence-shell electron configuration of the element just above E in the group is

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- (a) $3s^2 3p^4$
- (b) $3s^2 3p^5$
- (c) $3s^2 3p^6$
- (d) $3s^2 3p^3$

23. The electronic configurations of four elements are given below. Which element does not belong to the same family as the others?

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- (a) $[\text{Xe}] 4f^{14} 5d^{10} 6s^2$
- (b) $[\text{Kr}] 4d^{10} 5s^2$

(c) $[\text{Ne}] 3s^2 3p^5$

(d) $[\text{Ar}] 3d^{10} 4s^2$

24. The elements with atomic numbers 9, 17, 35, 53 and 85 belong to

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- (a) alkali metals
- (b) alkaline earth metals
- (c) halogens
- (d) noble gases

25. Which of the following pairs has both members from the same period of the periodic table?

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- (a) Na – Ca
- (b) Na – Cl
- (c) Ca – Cl
- (d) Cl – Br

26. The elements characterized by the outer electronic configuration ns^1 to $ns^2 np^6$ are collectively called

- (a) transition elements
- (b) representative elements
- (c) lanthanides
- (d) inner transition elements



27. f-Block elements are called inner transition elements because

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- (a) they have properties similar to those of transition elements
- (b) they exist in between transition elements
- (c) the last electron enters into the f-orbital of the penultimate shell
- (d) the last electron enters into any orbital of the penultimate shell

28. An element which belongs to the third period and group 16 of the periodic table has electronic configuration

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- (a) $1s^2 2s^2 2p^6 3s^2 3p^3$
- (b) $1s^2 2s^2 2p^6 3s^2 3p^4$
- (c) $1s^2 2s^2 2p^6 3s^2 3p^5$
- (d) $1s^2 2s^2 2p^4$

29. * In the long form of the periodic table, all the non-metals are placed in the

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- (a) s-block

(b) p-block

(c) f-block

(d) d-block

30. * Which period contains most of the man-made radioactive elements?

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- (a) Seventh
- (b) Fifth
- (c) Sixth
- (d) Both (a) and (c)

31. The electronic configuration of an element is $1s^2 2s^2 2p^6 3s^2 3p^3$. What is the atomic number of the element just below the above element in the periodic table?

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- (a) 33
- (b) 34
- (c) 36
- (d) 49

32. * An element X belongs to the fourth period and fifteenth group of the periodic table. Which one of the following is true regarding the outer electronic configuration of X? It has

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(a) partially filled d-orbitals and completely filled s-orbitals

(b) completely filled s-orbital and completely filled p-orbitals

(c) completely filled s-orbital and half-filled p-orbitals

(d) half-filled d-orbitals and completely filled s-orbitals

10 and 30 respectively, belong to the same period?

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(a) A, B, C

(b) B, C, D

(c) A, D, E

(d) B, D, E

33. * An element has electronic configuration $1s^2 2s^2 2p^6 3s^2 3p^4$. Its period, block and group are

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(a) Period 3, p-block, group 16

(b) Period 5, s-block, group 1

(c) Period 3, p-block, group 10

(d) Period 4, d-block, group 12

36. Which of the following elements forms a basic oxide?

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(a) Aluminium

(b) Carbon

(c) Nitrogen

(d) Magnesium

34. An element having atomic number 56 belongs to which block of the periodic table?

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(a) p-block

(b) d-block

(c) f-block

(d) s-block

37. Elements of which group form anions most readily?

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(a) Oxygen family

(b) Nitrogen family

(c) Halogens

(d) Alkali metals

35. Which of the given elements A, B, C, D and E, with atomic numbers 2, 3, 7,

38. Which of the following is non-metallic?

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(a) B

(b) Be

(c) Mg



(d) Al

39. The only non-metal which is liquid at ordinary temperature is

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- (a) Hg
- (b) Br
- (c) Na
- (d) None of these

40. Arrange the following elements in the order of increasing non-metallic character: Li, O, C, Be and F.

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- (a) $F < O < C < Be < Li$
- (b) $Li < Be < C < O < F$
- (c) $F < O < C < Be < Li$
- (d) $F < O < Be < C < Li$

